

Kushal - Profile

About Me

I am an Electronics, Communication and Information Engineering student at IOE, Pashchimanchal Campus, currently entering my 8th semester. I am interested in machine learning and artificial intelligence, with hands-on experience in Python-based data science tools. I also have front-end web development skills and a background in hardware design through PCB work.

Education

Bachelor's in Electronics, Communication and Information Engineering, Institute of Engineering (IOE), Pashchimanchal Campus (Western Regional Campus). Currently in 8th semester, with 7th semester exams completed.

Technical Skills

Programming and data science: Python, NumPy, pandas, Matplotlib, scikit-learn, C, C++.

Web development: HTML, CSS, JavaScript.

Database: SQL Server.

Hardware: Arduino, PCB designing.

Tools: MATLAB, used for visualizing signals such as amplitude modulation in communication systems coursework.

Extracurricular

Plays football, cricket, and table tennis.

Projects

Earthquake Disaster and Management System (group project, team of four). Built a system using two ESP32 microcontrollers, each paired with an accelerometer. The system required both accelerometers to detect motion above a threshold at the same time to confirm a real earthquake, reducing false alarms from a single sensor trigger. Earthquake magnitude was calculated from the accelerometer readings. Results were displayed on a dashboard, and an SMS alert was sent to a phone. My role was PCB designing and testing.

Student Performance Prediction (individual project). Built a K-Nearest Neighbors machine learning model to classify student academic performance as Poor, Average, or Excellent, based on study habits and related factors, using a dataset of over six thousand students. Used Random Forest to identify the most important features, then trained and tuned a KNN classifier, reaching about 78% accuracy. Published on GitHub with a full README.

Document Question-Answering Chatbot (individual project, in progress). Building a Retrieval-Augmented Generation (RAG) chatbot that answers questions from any PDF

document. The system loads and splits a PDF into chunks, converts the chunks into vector embeddings, retrieves the most relevant chunks for a given question, and uses a large language model to generate an answer grounded only in that retrieved content.